

View Reviews

Paper ID

2635

Paper Title

ClearEval: A Data-driven Benchmark for Evaluating Large Language Models in Tissue Optical Clearing

Reviewer #1

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines:

<https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

MIC

4. How would you describe the paper?

Methodological contribution

5. Please describe the main contribution of the paper.

The authors introduce a novel benchmark to assess the soundness of medical and biological knowledge in large language models by evaluating their ability to design a tissue optical clearing (TOC) protocol, a complex and application-driven task. They conduct experiments on state-of-the-art LLMs and further validate a subset of the results through comparison with human expert judgments.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

The central premise of the paper is compelling, suggesting that while LLMs have become highly capable at retrieving and articulating textual knowledge, gaps remain in their ability to apply this knowledge in complex, real-world scenarios. The dataset design is theoretically grounded, and the evaluation framework is substantive, moving beyond simplistic metrics such as BLEU. Additionally, the inclusion of human evaluation serves as an important sanity check for the findings.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

The work is presented in a meticulous and thorough manner. However, a few additional details would strengthen the paper: specifically, including the prompts used for both generation and evaluation in the LLM-as-judge phase would improve transparency and reproducibility. The overall motivation for the task could also be better articulated, particularly given that many state-of-the-art LLMs already demonstrate strong performance. Finally, providing more detail on the human evaluation protocol would enhance the credibility and interpretability of the results.

8. Please rate the clarity and organization of the paper.

Good

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The submission has provided an anonymized link to the source code, dataset, or any other dependencies.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

5. Accept — should be accepted, independent of rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

The core premise is compelling: a gap persists between knowledge acquisition and its application in complex settings. The benchmark design is thorough, and the evaluation framework is both rigorous and thoughtfully constructed. Overall, this represents a meaningful contribution toward assessing LLMs in medical and biological domains.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

Very confident (4)

21. [Post-rebuttal only] After reading the authors' rebuttal, please state your final opinion of the paper.

Accept

22. [Post-rebuttal only] Please justify your final decision from above.

I find the arguments convincing.

Questions

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MIC

4. How would you describe the paper?

Application study

5. Please describe the main contribution of the paper.

The authors evaluate 13 LLMs, 5 closed, 8 open, on the task of designing wet-lab protocols for Tissue Optical Clearing (TOC), a sample preparation technique for 3D microscopy. They contribute a dataset of 260 TOC protocols with expert annotations, and a multi-metric evaluation combining LLM-as-Judge scoring, rule-based checks for chemical incompatibility, and vector-similarity comparison to reference protocols. The headline findings are that all models exhibit two recurring failure modes on step durations and chemically incompatible reagent combinations.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

Task significance: protocol design is a difficult task requiring reasoning over reagents, timing, equipment, and mechanism. It is also underexplored relative to biomedical QA or literature retrieval, making it a worthwhile task for LLM evaluation.

Model coverage: the selection of 13 models (5 closed, 8 open) spans a reasonable breadth of the current frontier and open-weight landscape.

Actionable failure modes: identifying time estimation and chemical incompatibility as recurring failure modes is concrete and mechanistically interpretable, giving future work clear targets.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

Generalization: 260 protocols in a single subdomain (TOC) is insufficient to support general claims about LLM protocol-design capability. Comparable benchmarks operate at substantially larger scale and broader coverage, for example BioProBench with 27k protocols across 16 subfields.

Evaluation: 5 of the metrics rely on LLM-as-Judge without reported agreement against expert judgment on TOC, despite well-documented judge biases and agreeableness effects. The deterministic and vector-similarity checks cover only a narrow slice of protocol quality, leaving the evaluation stack's overall coverage thin.

Missing baselines: standard inference-time techniques known to materially affect reasoning performance were not evaluated, self-consistency, RAG or context over a protocol corpus, and tool use, and context windows/decoding parameters are not reported. These methods should be exhausted before claims on LLM capabilities are made.

8. Please rate the clarity and organization of the paper.

Satisfactory

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The submission does not provide sufficient information for reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

2. Reject — should be rejected, independent of rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

I recommend Reject because the work's central claims about LLM capability for protocol design are not supported by the evidence presented. The benchmark code and data at "<https://github.com/404NotFound422/ClearEval-public>" are not available. If they were, it is still confined to a single narrow subdomain at modest scale and the evaluation relies heavily on uncalibrated LLM-as-Judge without reported agreement with expert judgment, and standard prompting and tool-use baselines known to change reasoning performance substantially are absent. The curated TOC dataset and identified failure modes could be genuine contributions.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

Confident but not absolutely certain (3)

21. [Post-rebuttal only] After reading the authors' rebuttal, please state your final opinion of the paper.

Reject

22. [Post-rebuttal only] Please justify your final decision from above.

The rebuttal reframes the work as a TOC-specific pre-experimental checker, but the submission's abstract and discussion still make broader claims about a knowledge-execution gap in scientific AI that a single-subdomain benchmark cannot support. Clarifications on the two-stage design and dataset expansion are useful, but they do not

address the generalization concern that 260 protocols in one subdomain is modest relative to contemporary work such as BioProBench (27k protocols, 16 subfields). The RAG ablation is new experimental work absent from Section 5 and per meta-reviewer instructions should be set aside, leaving the missing-baselines critique intact. The repository README at <https://github.com/404NotFound422/ClearEval-public> is not in English, limiting reproducibility for the reviewing audience. My recommendation remains Reject.

Reviewer #3

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines:

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MIC

4. How would you describe the paper?

Methodological contribution

5. Please describe the main contribution of the paper.

The paper's main contribution is a data-driven benchmark for evaluating Tissue Optical Clearing wet-lab protocols generated by LLMs. More specifically, it proposes a benchmark that assesses protocols produced by models such as Claude, Chat-GPT etc. along two dimensions: their semantic coherence/description quality and qualitative benchmark criteria such as completeness, correctness, and effectiveness. A further contribution seems to be the introduction of an application index and an expert-informed evaluation logic intended to ground the assessment empirically, with the broader goal of moving beyond pure declarative knowledge retrieval toward more practically meaningful evaluation of scientific AI output.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

Interesting and relevant problem setting:

The paper addresses the evaluation of LLM-generated Tissue Optical Clearing wet-lab protocols, which is a timely and practically relevant problem at the intersection of scientific AI and laboratory protocol design.

Clear benchmark-oriented contribution:

A central strength is that the paper does not only discuss LLM outputs qualitatively, but proposes a data-driven benchmark for evaluating them in a more systematic way.

Good conceptual motivation:

The paper is built around a meaningful claim. A key bottleneck in scientific AI is not only knowledge retrieval but also the lack of empirical grounding. This gives the work a conceptual motivation.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

Insufficient related work coverage:

A major weakness is that the paper appears to discuss too little related work, and the references it does include focus mainly on general LLM weaknesses and evaluation, rather than on the actual application domain of Tissue Optical Clearing protocol design.

Unclear validity of the benchmark:

Although the benchmark idea is interesting, the paper apparently provides too little information about how the benchmark is actually constructed and about the quality of the expert knowledge behind it, making its validity hard to assess.

Insufficient explanation of expert alignment:

The paper should explain in much more detail how the expert alignment was performed and which aspects were collected from experts to build the benchmark. This is a central weakness because the benchmark's core claims depend on that alignment.

Unclear impact of the work:

Because the benchmark basis, expert alignment, and application-domain framing are not sufficiently established, the overall impact and significance of the research contribution remain unclear to me.

8. Please rate the clarity and organization of the paper.

Satisfactory

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The submission has provided an anonymized link to the source code, dataset, or any other dependencies.

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No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

3. Weak Reject — marginally below the acceptance threshold, but would not mind if accepted, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

The rating is based on the fact that the paper has an interesting core idea, but the central benchmark contribution is not sufficiently validated or documented to support acceptance in its current form. I find the topic timely and interesting: evaluating LLM-generated Tissue Optical Clearing wet-lab protocols with a data-driven benchmark is a meaningful direction, and the paper's attempt to move beyond generic LLM evaluation toward criteria such as semantic coherence, completeness, correctness, and effectiveness is clearly valuable. I also think the proposed application index and the broader data-centric evaluation idea are interesting and give the work some novelty.

However, the main reason for the weak reject is that the paper does not provide enough detail about the benchmark itself, especially about the expert alignment / human grounding on which its main claims seem to depend. It remains unclear how the experts were involved, which aspects were collected from them, and how exactly that knowledge was translated into the benchmark. Since this expert-grounded benchmark is the paper's central contribution, this lack of detail makes it difficult to judge whether the benchmark is actually valid, reliable, and reproducible. In that sense, the paper's own claim that the bottleneck in scientific AI is the lack of empirical grounding is undermined by the fact that this empirical grounding is itself not explained in a sufficiently transparent way.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

Somewhat confident (2)

21. [Post-rebuttal only] After reading the authors' rebuttal, please state your final opinion of the paper.

Reject

22. [Post-rebuttal only] Please justify your final decision from above.

The author feedback added some clarification, but for me that raises more questions (and curiosity) about the actual benchmark construction and metric validity. For example the selection process of 253 OEQs is still not transparent or understandable for me. Same for the QA-generation workflow for the MCQs. Since this data has huge influence on the overall performance of the checker more information on the data curation process is necessary.